

Towards interpretable artificial intelligence in clinical diagnosis

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Introduction

Bees, burger flippers, beer crafters. In the modern age, it's more difficult to think of what artificial intelligence (AI) can't do than to think of what it can. Its applications are being explored in a wide variety of fields, and healthcare is no exception. Valued for their efficiency, low cost, and availability, AI models such as neural networks are gaining popularity as tools to assist medical professionals in clinical diagnoses. Over the past few years, the development and adoption of such models has resulted in many questions regarding what must be done to ensure ethical usage. The lack of transparency commonly associated with computer intelligence has resulted in a strong demand for explainable AI (XAI).

Existing ethical standards have made the push for XAI particularly prevalent in healthcare (Amann et al, 2020). Historically, the field of medical ethics has mostly considered direct patient-provider interactions; regarding clinical diagnosis, an example of this interaction can arise if a patient wants an explanation for their diagnosis. Transparency in this explanation is crucial to maintain patient trust in medical professionals, and opacity in such a situation would violate principles of medical ethics, most notably patient autonomy (Farhud & Zokaei, 2021). Patient dependence on healthcare providers creates a moral obligation in providers to ensure patient well-being, a duty that would be breached by leaving them to the mercies of a machine. XAI helps balance technological advancement with ethical usage and care.

Unlike simpler machine learning models, neural networks are not inherently interpretable and thus have developed a reputation as "black-box" decision makers. Various interpretability methods have been developed to try and solve this issue; medical image classification tasks have been the subject of numerous tests (Borys et al, 2023). These methods can generally be divided into two categories: perturbation-based and gradient-based. The first type makes systematic modifications to input features; through observing the resulting changes in the output, a local

approximation of the decision boundary is formed. The second type examines the network itself to understand the sensitivity of the output with respect to each input feature, values known as gradients. This is commonly done using backpropagation, an algorithm that computes these gradients backwards, starting with the output and working through each hidden layer.

When considering the practical applications of models like the one shown here (i.e., actually using them in clinical diagnosis), the possibility of XAI methods inaccurately capturing the most important features presents an important issue. A model that fails to correctly explain its reasoning would likely be faced with distrust by a patient. As a result, the explanations generated by any XAI method should be assessed through various metrics to gauge confidence in any given model. This in turn would allow wider ethical usage, as the utilization of flawed models in diagnosis would not comply with existing quality standards in healthcare.

One way to analyze explainability results is to examine consistency. If multiple methods consistently identify certain features as important, it may indicate greater confidence in the true impact of those features and in the method itself to successfully accomplish its goal. Here, the use of consistency as an informative aspect of interpretability is explored.

Methodology

Four popular XAI methods for image classification tasks are demonstrated here. Randomized Input Sampling for Explanation (Petsiuk et al, 2018), abbreviated RISE, is a method that randomly perturbs input images and uses the resulting model output to estimate feature contribution. A second method, integrated gradients (IG) is gradient-based and considers a path from a baseline (such as a black image with each pixel set to 0) to the input data. Multiple gradients are calculated along and then integrated over this path, allowing for a continuous picture of each feature's contribution. Guided backpropagation is similarly gradient-based, and focuses on positive contributions to the model's prediction by suppressing negative values. Finally, gradient-based saliency maps use similar gradients calculations to those of guided backpropagation, but account for all gradients by applying absolute value.

The dataset contains images of brain tumors obtained via magnetic resonance imaging (MRI). Each image belongs to one of four classes: glioma, meningioma, pituitary, or no tumor. This set contained images from multiple sources, gathered by Kaggle (Nickparvar, 2021).

The model employed the EfficientNetV2S architecture from TensorFlow, followed by dropout with strength 0.25, a fully-connected layer with 128 neurons, batch normalization, relu activation, dropout with strength 0.3, and a final fully-connected layer with 4 neurons.

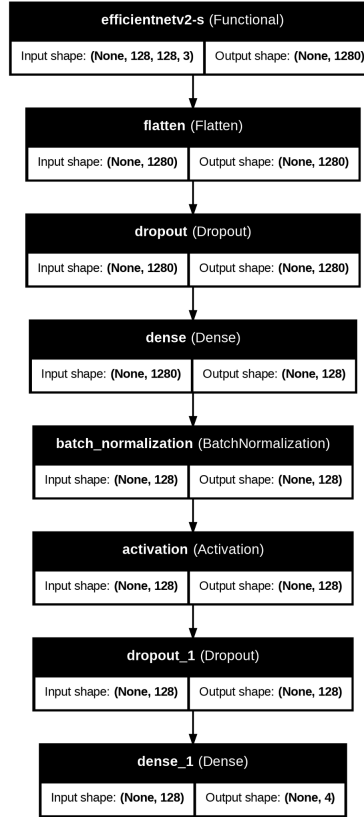


Figure 1. The architecture of the demonstrated model

All attribution methods can be visualized with saliency maps, which highlight the features identified as important by the respective method. However, visual assessment alone is considered insufficient to understand the true quality of any XAI method (Adebayo et al, 2018). To better analyze if the features that most contribute to an image's classification were correctly identified, an ablation study was conducted, a technique used to assess the quality of XAI methods (Hameed et al, 2022). Ideally, model performance should drop off steeply as the first few features are removed; if this is not the case, it indicates that the method may not be accurately capturing the most salient features.

To analyze consistency, the agreement between pixel importance values following the application of XAI methods was explored. The ablation process involved ranking the pixels by importance, allowing features to be ablated in an order. Higher agreement between the rankings

produced by different methods may indicate more consistent attribution of a model's decision to specific pixels in an image. This in turn may give an idea of the XAI methods that most accurately identify the contributing features of any given input data.

Results

The model achieved a maximum accuracy of around 98%. Following training and testing, the aforementioned XAI methods were run on the model using a subset of the testing data. Importance maps were created for each sample, as demonstrated in Fig. 2.

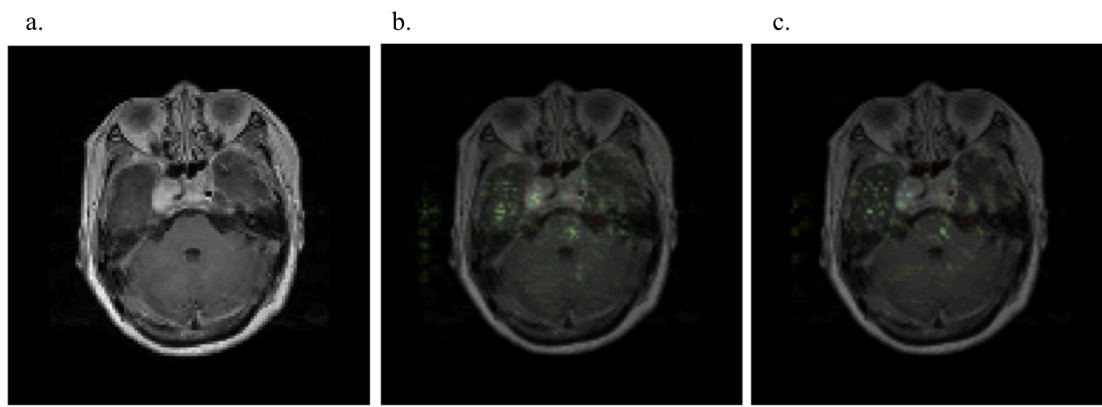


Figure 2. Examples of the saliency maps for an image (2a) generated by each method, with 2b and 2c corresponding to gradient-based saliency maps and guided backpropagation, respectively.

Visual examination is not particularly informative and reveals limited information.

Saliency maps, such as those shown above, may not necessarily be the most insightful. For instance, the maps do not focus solely on the imaging section; in other words, the important features are shown to be on the brain section and not on the background. This makes intuitive sense, as the solid background is unlikely to be informing the model's decision, but this limits what can be deduced regarding the decision-making process. This reinforces the idea that purely visual examination is insufficient to extract insights from XAI.

The ablation study removed features in increments of 10%, assessing model performance at each point. The results are shown in Fig. 3.

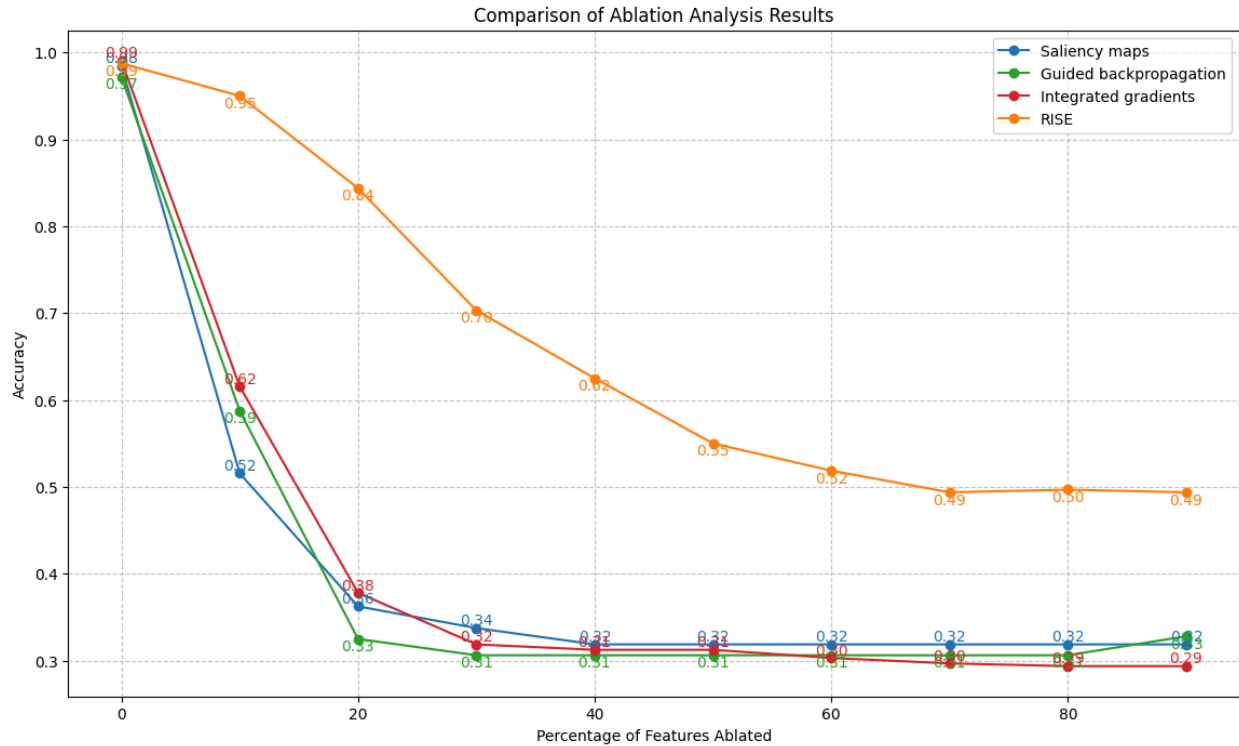


Figure 3. Performance drops off at a decreasing rate as features are ablated, indicating that the methods may be accurately capturing important features. There is also a significant difference between RISE and the other methods, likely resulting from differences in calculation (perturbation versus gradient).

To assess consistency, the pixel-wise agreement between the two methods was examined. Because the pixel rankings produced by each method were used as a comparison, three rank-based measures were employed: Spearman correlation, Kendall's tau, and normalized discounted cumulative gain (NDCG). These values were calculated for each pair of attribution methods for each image within a batch of 32 and averaged.

	Method1	Method2	Rank Correlation	Kendall's Tau	NDCG Score
0	guided_backpropagation	rise	-0.000505	-0.000391	0.936603
1	integrated_gradients	guided_backpropagation	-0.001523	-0.001144	0.935439
2	integrated_gradients	rise	0.002364	0.001583	0.935640
3	integrated_gradients	saliency	-0.002076	-0.001659	0.935411
4	saliency	guided_backpropagation	0.282834	0.274981	0.974605
5	saliency	rise	-0.026414	-0.018060	0.934200

Table 1. Quantification of similarity between the rankings produced by various attribution methods across one batch of 32 images. Despite lower values of Spearman correlation and Kendall’s tau, the high NDCG scores indicate that the methods may be labeling similar features as the most important.

Discussion

All methods, as well as all XAI methods, have varying strengths and weaknesses. In using the agreement metric, the goal is to reduce the possibility of misinformative saliency maps. While consistency isn’t guaranteed to be a true measurement of accuracy, this repeated validation can increase the trust of patients in AI models for diagnosis.

The results of the consistency test show notable gaps between the first two and the last measure, which likely results from the difference in how NDCG is calculated. The image dimensions (128 x 128) mean that the rankings contain thousands of pixels, and of the three methods, only NDCG gives greater weight to the features with the greatest importance scores. This indicates that while methods may not produce an overall similar feature ranking, the features considered most salient for the model’s prediction display at least some similarity across methods.

This study, particularly the agreement metrics, is limited by the scope of data and methodology tested here. The lack of pixel-level ground truth can make it hard to ascertain the relationship between consistency and accuracy. As such, use of this metric should be done in conjunction with various other XAI testing methods, such as Remove and Retrain (Hooker et al, 2018). Furthermore, evaluation of the metric on a variety of models and datasets, including those of varying modalities, can improve confidence in its results.

Proper use of XAI methods has the potential to improve both confidence and fairness. Forcing a model to justify its prediction can result in generalizable insights, which can aid current diagnostic methods. However, it is still important to remain vigilant of XAI's shortcomings. Healthcare professionals must leverage their own expertise to prevent the spread of model-derived misinformation. Given current limitations in regulatory standards, data quality, and patient trust, the professionally trained human mind remains the gold standard of clinical diagnosis.

XAI also helps resolve a major issue across all data-driven models: bias. Flaws in training data are inevitable, as the data-gathering process is prone to human error. These limitations can manifest in the model's outputs, which is an issue when looking to use these models in diagnosis. The theoretical objectivity of a computer is one of its most attractive aspects, but clearly this is not demonstrated in practice. This only furthers the need for XAI, as attribution methods can help pinpoint sources of bias that a person would not be able to identify nor address (Vainio-Pekka et al, 2023).

When considering the capabilities of AI in clinical diagnosis, it is critical to weigh both benefits and drawbacks. Given the severity of potential consequences of AI misuse in diagnoses, it is necessary to subject any decision, no matter who or what it is made by, to rigorous standards. With the goal of improving both quality and availability of healthcare, utilization of AI is almost certainly necessary, but ethical values must be maintained as well. Explainability is key to this.

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